Lesson 08.4:

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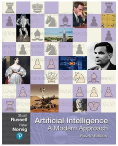
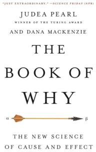
References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017
- Facure, Causal Inference in Python, 2023





Sales example

- There are stores that put product on sales for Christmas

	store	weeks_to_xmas	avg_week_sales	is_on_sale	weekly_amount_sold
0	1	3	12.98	1.0	219.60
1	1	2	12.98	1.0	184.70
2	1	1	12.98	1.0	145.75
3	1	0	12.98	0.0	102.45
4	2	3	19.92	0.0	103.22
5	2	2	19.92	0.0	53.73

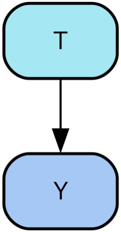
  where

  - `store`: id of the store
  - `weeks_to_xmas`: how many weeks before Christmas the measurement was done
  - `avg_week_sales`: how much they sold in average
  - `is_on_sale`: if store puts product on sale
  - `weekly_amount_on_sale`: how much that store has sold in that week

- Causal inference question: *"what is the impact of T on Y?"*
  - E.g., what is the impact of `is_on_sale` on `weekly_amount_sold`?
  - E.g., when is the best moment to have sales (i.e., `weeks_to_xmas`)?

Definition

- **Treatment** T
  - $T_i = 1$ if unit i has received the treatment
  - Sometimes T is denoted with D
  - E.g., `is_on_sale` at `weeks_to_xmas`, `is_on_sale`, `received_discount`, `saw_ad`
- **Outcome** Y
  - The variable one wants to influence, Y_i
  - E.g., `weekly_amount_sold`
- **Unit of analysis**
  - The entity for which outcomes are measured

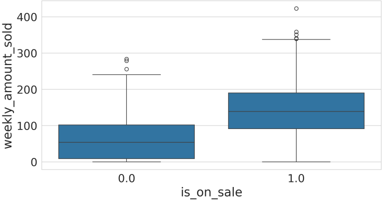


Study	**Unit of analysis**
Drug experiment	Patient
Marketing experiment	Customer
Sales promotion	Store or product
Education policy	Student

The Fundamental Problem of Causal Inference

- **The Fundamental Problem of Causal Inference**
  - *"You can never observe the same unit with and without treatment"*

- **Example**
  - Plot outcome `weekly_amount_sold` conditioned on treatment `is_on_sale`
  - Stores with price drops sell more (from 50 to 150)
  - Matches intuition: *"people buy more when prices are low due to sales"*

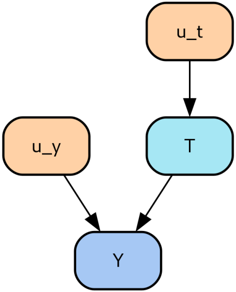


- **Is it true?**
  - Maybe only larger businesses afford aggressive price drops?
  - Maybe customers buy close to Christmas anyway?
- Observing both counterfactual situations would reveal the effect of price drops (this is **impossible**)

Causal Model

- A **causal model** is a series of assignments $\leftarrow$
  - Causality is non-reversible: you can't rearrange the equation
- **Example**

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$



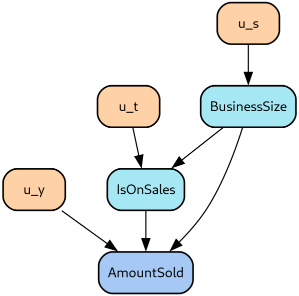
where:

- u_t, u_y are *exogenous* variables (outside the model)
  - ∗ Account for everything not covered by model variables
- Treatment T and u_y cause outcome Y
- T and Y are *endogenous* variables (inside the model)

Causal Model: Sales Example

- Causal model for sales example:

$$\begin{cases} BusinessSize \leftarrow f_s(u_s) \\ IsOnSales \leftarrow f_t(BusinessSize, u_t) \\ AmountSold \leftarrow f_y(IsOnSales, BusinessSize, u_y) \end{cases}$$



- One can make an assumption of linear model for $f_y(.)$ and say

$$AmountSold = \alpha + \beta_1 IsOnSales + \beta_2 BusinessSize + \varepsilon$$

Applying a Treatment to Sales Example

- Assume the **causal model** is:

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

  - Ask *"what would happen to the outcome Y if I were to set the treatment to t_0?"*

$$\begin{cases} T \leftarrow t_0 \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

  - The intervention is represented by the $do()$ operator, e.g., $do(T = t_0)$
- Note that **conditioning** and **intervention** are different:

$$\mathbb{E}[AmountSold | isOnSales = 1] \neq \mathbb{E}[AmountSold | do(IsOnSales = 1)]$$

  - The **first** is what happens to businesses that make a price cut
    - Some business (e.g., big businesses) are "self-selecting" to cut prices
  - The **second** is what happens if you force every business to engage in sales

Individual Treatment Effect

- Express the impact of the treatment T on the outcome Y for an individual unit i

$$\tau_i = Y_i | do(T = t_1) - Y_i | do(T = t_0)$$

  - The effect τ_i of going from treatment t_0 to t_1 for unit i is the difference in the outcome of that unit under t_1 compared to t_0
  - E.g., $AmountSold_i | do(IsOnSales = 1) - AmountSold_i | do(IsOnSales = 0)$
- You can only observe one term due to the fundamental problem of causal inference
  - Represent it in theory, but can't recover it from data

Potential Outcomes

- Aka "counterfactuals"
- The *do*() operator represents *"unit i outcome is Y if treatment is set to t"*

$$Y_{t,i} \triangleq Y_i | do(T_i = t)$$

  - The first is "Rubin's notation", the second is "Pearl's notation"
- For binary treatment use Y_{0i} (without treatment) vs Y_{1i} (with treatment)
  - One is "factual" (i.e., observable)
  - The other is "counterfactual" (i.e., theoretical, not observable)
- There is a slight difference
  - $Y_t = Y | do(T = t)$ is "what a business would sell if it cut prices"
  - $Y | T = t$ is "what a business that cut the prices sold"

Consistency of the Treatment

- There is a single version of the treatment
  - I.e., $T = T_i$ means only one thing
- E.g.,
  - A drug can come in multiple dosages: only one is used
  - Giving a discount coupon is treated as the value of the coupon is the same, it's given only once, in a way that is always equivalent
  - In "help from a financial planner", is "help" a one-time consultation or regular advice?

No Interference

- Aka “Stable Unit of Treatment Value (SUTVA)”
- The effect of one unit is not influenced by the treatment of other units
  - E.g., there are no spillovers or network effects
- E.g., vaccinating one person makes other close people less likely to catch the illness even if they don’t get the treatment

Average Treatment Effects

- Assume you can see both potential outcomes Y_{1i} and Y_{0i} at the same time (which is impossible)
- **Average Treatment Effect (ATE)**
  - The impact that the treatment T would have on the average:

$$ATE = \mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

- E.g., average impact of price cuts on amount sold:

$$ATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i}]$$

Conditional Average Treatment Effects (CATE)

- The impact of the treatment on the units in a group defined by $X = x$

$$CATE = \mathbb{E}[(Y_{1i} - Y_{0i})|X = x]$$

- E.g., the impact of having sales during the week of Christmas:

$$CATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i}|weeksToXmas = 0]$$

- Like ATE but on a subgroup with a certain characteristics

Average Treatment effect on the Treated (ATT)

- The impact of the treatment on the units that got the treatment:

$$ATT = \mathbb{E}[(Y_{1i} - Y_{0i})|T = 1]$$

- E.g., how much sales increase for the business engaged in price cuts:

$$ATT = \mathbb{E}[(AmountSold_{1i} - AmountSold_{0i})|IsOnSales = 1]$$

- We can define these quantities ATE, $CATE$, ATT, but not **measure them** directly
  - Because we can't observe both potential outcomes Y_{1i} and Y_{0i} at the same time

Bias

- **Bias** is what makes association different from causation
- The claim *"cutting prices increases the amount sold by a business"* can be questioned with:
  - *"Those businesses that cut prices could probably have sold more anyway, even with no price cuts"*
- More formally the argument refers to potential outcomes
  - We are trying to estimate $\mathbb{E}[Y_{it}] \triangleq \mathbb{E}[Y_i | do(T = t)]$ using $\mathbb{E}[Y_i | T = t]$ but this is a "biased estimator" (i.e., an incorrect one)

The Bias Equation

- **Causal inference** is interested in:

$$\mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y|do(T=1)] - \mathbb{E}[Y|do(T=0)]$$

- **What is measured** is:

$$\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0]$$

- Let's express one in terms of the other
  - Replacing the observed outcomes (each unit has two potential outcomes but only one can be observed and the other is 0)

$$Y|T=1 = (Y_1|T=1) + \cancel{(Y_0|T=1)} = Y_1|T=1$$

  - Adding and subtracting $\mathbb{E}[Y_0|T=1]$, reordering and merging expectations

$$\mathbb{E}[Y|T=1] - E[Y|T=0] = \mathbb{E}[Y_1 - Y_0|T=1] + \underbrace{\mathbb{E}[Y_0|T=1] - \mathbb{E}[Y_0|T=0]}_{\text{Bias}}$$

  - Association = causation + a bias term

The Bias Equation: Remarks

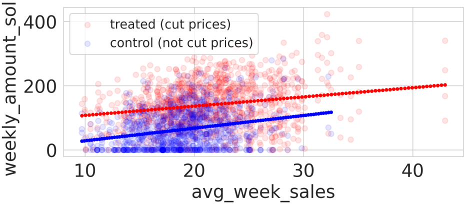
- Association = causation + a bias term

$$\mathbb{E}[Y|T=1] - E[Y|T=0] = \mathbb{E}[Y_1 - Y_0|T=1] + \underbrace{\mathbb{E}[Y_0|T=1] - \mathbb{E}[Y_0|T=0]}_{\text{Bias}}$$

- Bias arises from differences between treated and control groups (besides the treatment)
  - Unobservable factors change together with treatment
    - E.g., there are confounders
  - E.g., treated and untreated businesses differ beyond price cuts
    - E.g., differ in size, location, day of the week on which they have a sale, cities they are located in
- For association to be causation, treated and control units must be **exchangeable**

The Bias Equation: Visual Example

- Plot `weekly_amount_sold` by business size (`avg_week_sales`)
  - Color points: treatment (red) / control (blue)
  - Treated units concentrated on the right



- Difference in amount sold between groups has two causes
  - *Treatment effect*: Price cut increases amount sold
  - *Business size*: Bigger businesses sell more, can do more price cuts
- The challenge for causal inference is to untangle the two effects